

Md. Robiul Islam Roman

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@ <https://roman-rir.github.io/Portfolio/>



PROFESSIONAL SUMMARY

Electrical and Electronic Engineering graduate from AIUB, currently pursuing a Master of Engineering (M.Engg.) in Computer Science & Engineering at BRAC University. Proficient in MATLAB, Python, C/C++, and SystemVerilog, with hands-on experience in real-time data acquisition, biomedical signal processing, RTL/VLSI design, and embedded systems development.

SKILLS

Programming Languages:	PLC, MATLAB, C/C++, Python, Javascript, SystemVerilog
EDA & Design Tools:	Cadence Virtuoso, Xcelium, NI Multisim, CST Microwave Studio, Proteus, CAD, SolidWorks
Hardware & Embedded:	Raspberry Pi, FPGA, Embedded Systems, STM32
Generative AI & Backend:	<i>n8n, Zapier, AI agents, Fast API, REST APIs</i>
AI & ML:	CNN, Logistic Regression, SVM, Decision Trees, KNN, LSTM, RNN
Operating Systems & Version Control:	Command-Line Interface (CLI), Git, GitHub, Windows, Linux

EDUCATION

Master of Engineering (M.Engg.) — Computer Science & Engineering *April 2025 – Present*
BRAC University

Bachelor of Science (BSc) — Electrical and Electronic Engineering *Graduated February 2026*
American International University–Bangladesh (AIUB)
CGPA: 3.74 / 4.00

Higher Secondary Certificate (HSC) — Science *January 2021*
Kurmitola High School and College, Dhaka, Bangladesh

EXPERIENCE

AI Automation Engineer *Jul 2026 – Present*
Softvence Agency Alpha, Dhaka

- Designed and maintained AI automation workflows using n8n and GoHighLevel (GHL).
- Built AI-powered chatbots and automated business processes by integrating LLM APIs, REST APIs, and webhooks.
- Optimized automation workflows to improve operational efficiency and client engagement.

Industrial Attachment Trainee *Nov 2025 – Dec 2025*
Energypac Power Generation PLC, Dhaka

- Completed hands-on industrial training on diesel generators and Automatic Transfer Switch (ATS) systems.
- Gained practical knowledge of generator operation, load handling, and ATS functionality in large-scale industrial power environments.

CERTIFICATIONS & AWARDS

Best Capstone Project Award — Faculty of Engineering, AIUB *October 2025*

- Led a team developing a Continuous Cuffless Blood Pressure Monitoring System, coordinating hardware design, embedded software, sensor integration, and testing phases.
- Showcased the project at Broadband Expo 2026 (ISPAB) and an international symposium hosted by the Magna

Charta Observatory. **Runner-Up on Poster Competition** — ROBOLUTION 2025 (November 2025)

Dean's List Honors — Faculty of Engineering, AIUB

- Awarded for three consecutive semesters: Fall 2022 (Mar 2023), Spring 2022–23 (Feb 2024), Summer 2024–25 (Oct 2025).

Academic Scholarship — Faculty of Engineering, AIUB

- Awarded academic scholarship for outstanding academic performance from **Spring 2022–23 to Spring 2024–25**.

RTL Design, Verification, Synthesis & PnR for Digital VLSI Design — AIUB Institute of Continuing Education (ICE)

- Hands-on training covering the complete digital VLSI design flow: RTL coding, functional verification, synthesis, STA, and physical design (PnR) using industry-standard EDA tools.

Chip-Based VLSI Design for Industrial Applications — Coursera & L&T EduTech (October 2025)

- Completed specialization covering Digital Design, VLSI Chip Design (Electric VLSI EDA), VHDL Programming, and FPGA Architecture.

Data Analysis with Python — Institute of Information Technology (IIT), Jahangirnagar University

- Completed professional training in Python-based data analysis under the EDGE Project, Bangladesh Computer Council, ICT Division (**Nov 2024 – Jan 2025**).

PROJECTS

RTL design, synthesis, and place-and-route implementation of a RISC processor. | [GitHub](#)

- Designed pipelined Developed RTL design, synthesis flow, and place-and-route implementation of a RISC-V3 processor using SystemVerilog.

Cuffless Blood Pressure Estimation using ECG & PPG Signals | [GitHub](#)

- Developed a portable, wearable device for continuous non-invasive blood pressure estimation using ECG and PPG signal fusion.

Raspberry Pi-Based Person Tracking Robot | [GitHub](#)

- Built a robot with real-time face/person detection and autonomous following capability using computer vision on embedded hardware.

Physical Layout Design using Cadence Virtuoso

- Implemented and verified physical IC layouts using the Cadence Virtuoso toolchain, including DRC and LVS checks.

MATLAB-Based Music Signal Feature Extraction

- Developed a MATLAB pipeline for extracting time-domain and frequency-domain audio features for classification tasks.

PUBLICATIONS

Development of a PPG-Based Low-Cost Cuffless Continuous Blood Pressure Monitoring with Vital Sign Detection System

- IEEE BECITHCON, November 2025.

Predicting Diabetes in Bangladesh Using Machine Learning: A Data-Driven Approach

- International Journal of Modern Engineering and Management (IJMEM), Maharashtra, India, Vol. 2, Issue 4, April 2025.

Portable Health Insights: Designing a BIA-Enabled Smartwatch for Clinical and Personal Use

- RURS 1st International Student Conference and Research Fair 2024, University of Rajshahi, February 22–23, 2025.

REFERENCES

Dr. Shahriyar Masud Rizvi

Associate Professor, Department of EEE, AIUB

Deputy Director, Center for VLSI and Embedded Systems (CVES)

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Dr. Md. Hasan Imam

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